

Shuheï Watanabe

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Education

- Oct 2020 – Oct 2023 **University of Freiburg**, Freiburg, Germany.
MSc in Computer Science. Supervisor: Prof. Frank Hutter.
Overall GPA: 1.1/5.0 (1.0 is the best grade).
Thesis: Significant Runtime Reduction for Asynchronous Multi-Fidelity Optimization on Zero-Cost Benchmarks.
- Apr 2016 – Mar 2020 **The University of Tokyo**, Tokyo, Japan.
BSc in Systems Innovation. **Ranked 1st** out of 37 students by GPA.
- Apr 2014 – Mar 2016 **The University of Tokyo**, Tokyo, Japan.
Bachelor of College of Arts and Science, Natural Science 1.

Employment

- Jan 2026 – Present **SB Intuitions Corp.**, Tokyo, Japan. (Robotics Senior Research Scientist)
Involved in cloth folding demo project.
- Jun 2024 – Dec 2025 **National Institute of Advanced Industrial Science and Technology (AIST)**, Tokyo, Japan. (Part Time Visiting Researcher at Social Intelligence Research (SIR) Team)
Mentored Chisa Mori & Kaichi Irie (see below for more details).
- Oct 2023 – Dec 2025 **Preferred Networks Inc.**, Tokyo, Japan. (Full Time Research Engineer)
Core Optuna developer. Delivered significant speedup of TPESampler (300x), the default sampler in Optuna, and GPSampler (10x faster than BoTorch). Led the development of GPSampler and its extensions. Worked on Deployed Optuna in materials & physics simulation applications.
- Dec 2020 – Sep 2023 **Machine Learning Lab**, Freiburg, Germany.
Development of Auto-PyTorch, an AutoML tool.
- Sep 2018 – Sep 2020 **AIST**, Tokyo, Japan. (Full Time Technical Staff at SIR Team)
Co-first authored “Warm Starting CMA-ES for Hyperparameter Optimization” (AAAI’21). Conducted a large-scale experiment on a cluster.

Awards / Honors

- Aug 2026 **KDD Best Reviewer** (Research Track), ACM SIGKDD 2026.
- Sep 2023 **AutoML 2023 Travel Awards** (500 EUR).
- Aug 2023 **IJCAI-AIJ 2023 Travel and Accessibility Grant Program** (1,000 USD).
- Oct 2022 **NeurIPS 2022 Complimentary Registration** (350 USD).
- Oct 2022 **ELIZA MSc Scholarship** (12,000 EUR, 4 students selected from the University).
- Oct 2022 **Deutschlandstipendium** (3,600 EUR).

Jul 2022	1st Prize in AutoML2022: Multiobjective Hyperparameter Optimization for Transformers
Sep 2020	ITO Foundation for International Education Exchange (48,000 USD, AR: 6.7%).
Mar 2020	Hatakeyama Award from the Japan Society of Mechanical Engineers. Awarded for the distinctive grades at the University of Tokyo (AR: 5/340=1.5%).
May 2019	PRMU 2018 Yearly Research Encouragement Award. For “ <i>Speed Up of Hyper-Parameter Tuning with Nelder-Mead Method by Parallel Computing</i> ” (AR: 3/170=1.8%).

Publications & Talks

AR, ○, and ♣ refer to the acceptance rate, the presenter and the equally contributed authors, respectively. The total citations on Google Scholar is **1800+** as of October 2026.

Refereed Conference Publications

1. ○ K. Baba, Y. Ozaki, **S. Watanabe** (2026). Conditional PED-ANOVA: Hyperparameter Importance in Hierarchical & Dynamic Search Spaces. ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD).
2. ○ C. Mori, **S. Watanabe**, M. Onishi, T. Itoh (2025). User Preference-Based Parallel Coordinate Plots: Its Application in Guidance Planning. International Conference on Pedestrian and Evacuation Dynamics (PED).
3. ○ ♣ C. Mori, ♣ **S. Watanabe**, M. Onishi, T. Itoh (2025). Preference-Optimal Multi-Metric Weighting for Parallel Coordinate Plots. International Conference Information Visualisation (iV).
4. ○ **S. Watanabe**, N. Mallik, E. Bergman, F. Hutter (2024). Fast Benchmarking of Asynchronous Multi-Fidelity Optimization on Zero-Cost Benchmarks. AutoML Conference. Preliminary: AutoML Conference Workshop Track 2023.
5. ○ **S. Watanabe**, F. Hutter (2023). c-TPE: Tree-Structured Parzen Estimator with Inequality Constraints for Expensive Hyperparameter Optimization. International Joint Conference on Artificial Intelligence (IJCAI) (AR: 644/4566≈14%). Preliminary: GPSMDM Workshop at NeurIPS 2022.
6. ○ **S. Watanabe**, N. Awad, M. Onishi, F. Hutter (2023). Speeding Up Multi-Objective Hyperparameter Optimization by Task Similarity-Based Meta-Learning for the Tree-Structured Parzen Estimator. International Joint Conference on Artificial Intelligence (IJCAI) (AR: 644/4566≈14%). Preliminary: MetaLearn Workshop at NeurIPS 2022.
7. ○ **S. Watanabe**, A. Bansal, F. Hutter (2023). PED-ANOVA: Efficiently Quantifying Hyperparameter Importance in Arbitrary Subspaces. International Joint Conference on Artificial Intelligence (IJCAI) (AR: 644/4566≈14%). (**The default algorithm in Optuna**)
8. ○ S. Shigenaka, S. Takami, **S. Watanabe**, Y. Tanigaki, Y. Ozaki, M. Onishi (2021). MAS-Bench: Parameter Optimization Benchmark for Multi-Agent Crowd Simulation. International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS).

9. ○ ♣ M. Nomura, ♣ **S. Watanabe**, Y. Akimoto, Y. Ozaki, M. Onishi (2021). Warm Starting CMA-ES for Hyperparameter Optimization. AAAI Conference on Artificial Intelligence (AAAI). (AR: 1692/9034=19%). Preliminary: MetaLearn Workshop at NeurIPS 2019.
10. ○ S. Takenaga, **S. Watanabe**, M. Nomura, Y. Ozaki, M. Onishi, H. Habe (2020). Evaluating Initialization of Nelder–Mead Method for Hyperparameter Optimization in Deep Learning. International Conference on Pattern Recognition (ICPR). Oral Presentation.
11. ○ Y. Ozaki, Y. Tanigaki, **S. Watanabe**, M. Onishi (2020). Multiobjective Tree-Structured Parzen Estimator for Computationally Expensive Optimization Problems. The Genetic and Evolutionary Computation Conference (GECCO).
12. ○ **S. Watanabe**, Y. Ozaki, Y. Bando, M. Onishi (2019). Speeding Up of the Nelder–Mead Method by Data–Driven Speculative Execution. Asian Conference on Pattern Recognition (ACPR). Oral Presentation. (**Oral presentation: 36/273=13%**)

Refereed Journal Publications

1. ♣ Y. Ozaki, ♣ **S. Watanabe**, T. Yanase (2026). OptunaHub: A Platform for Black-Box Optimization. Journal of Machine Learning Research (JMLR).
2. ○ S. Shigenaka, S. Takami, **S. Watanabe**, Y. Tanigaki, M. Onishi (2024). MAS-Bench: A Benchmarking for Parameter Calibration of Multi-Agent Crowd Simulation. Journal of Computational Social Science.
3. ○ Y. Ozaki, Y. Tanigaki, **S. Watanabe**, M. Nomura, M. Onishi (2022). Multiobjective Tree-Structured Parzen Estimator. Journal of Artificial Intelligence Research (JAIR).

Selected Non-Archival Publications

1. ○ ♣ **S. Watanabe**, ♣ K. Baba (2027). No Sigmoid Required: Exact Feasibility for Constrained q EHVI. Under review at International Conference on Learning Representations (ICLR).
2. ○ K. Irie, **S. Watanabe**, M. Onishi (2026). Batch Acquisition Function Evaluations and Decouple Optimizer Updates for Faster Bayesian Optimization. Workshop on AI to Accelerate Science and Engineering (AI2ASE) at AAAI.
3. **S. Watanabe** (2023). Tree-Structured Parzen Estimator: Understanding Its Algorithm Components and Their Roles for Better Empirical Performance. arXiv:2304.11127. (**900+ citations on Google Scholar**, the first detailed paper about the Optuna default sampler)
4. ○ Y. Ozaki, ○ **S. Watanabe**, M. Onishi (2019). Accelerating the Nelder–Mead Method with Predictive Evaluation. Workshop on Automated Machine Learning (AutoML) at ICML.
5. ○ **S. Watanabe**, Y. Ozaki, M. Onishi (2019). Speed Up of Hyper-Parameter Tuning with Nelder–Mead Method by Parallel Computing. Pattern Recognition and Media Understanding (PRMU). **PRMU 2018 Yearly Research Encouragement Award** (AR: 3/170=1.8%).

Talks

1. ○ **S. Watanabe** (2026). LLM Meets Optuna: Potential, Challenges, and Lessons. Meta Adaptive Experimentation Workshop. (Invited)

2. ○ ♣ **S. Watanabe**, ○ ♣ Y. Ozaki (2025). Future Directions of AutoML from the Optuna View. NII Shonan Meeting (Advancing Automated Machine Learning). (Invited)
3. ○ **S. Watanabe**, H. Imamura, C. Shinagawa, K. Shinohara, S. Takamoto, J. Li (2024). Multi-Objective Bayesian Optimization for Materials Discovery with Neural Network Potential – An Application to Li-Ion Battery Cathode Material. Materials Research Society Fall Meeting & Exhibit.

Academic Service

Reviewer

- International Conference on Learning Representations (2027).
- ACM SIGKDD Conference on Knowledge Discovery & Data Mining (2026). Recognized as a **KDD Best Reviewer** (Research Track).
- AutoML Conference (2023).

Mentoring & Supervision

Jun 2024 – Mar 2026	Chisa Mori , MSc Student, AIST. Parallel coordinate plots for multi-objective problems (Accepted to iV).
Jul 2024 – Present	Kaito Baba , MSc Student, Preferred Networks Inc. 1. Development of constrained optimization for the Gaussian process-based sampler (single-objective, multi-objective). 2. PED-ANOVA for conditional search spaces (Accepted to KDD).
Aug 2025 – Present	Kaichi Irie , MSc Student, Preferred Networks Inc. & AIST. Development of parallel processing in the Gaussian process-based sampler (article, workshop paper).

Technical Highlights

- Coding-Agent Management (token efficiency tracking)
- Applications of Bayesian Optimization (materials science & engineering design)
- Physics-Based Simulation (lattice Boltzmann, numerical integration, constraint satisfaction)
- Cluster Experience (MPI, parallel programming, large-scale experiments, MOAB, Slurm)
- Software Engineering (Python, C++, PyTorch, team development)

Miscellaneous

- **Japanese** (Native Language), **English** (C1, TOEFL iBT: 100), **German** (B2)
- Approx. Top 3.5% (highest) algorithm programmer in **AtCoder** mostly using C++.